

EEG Cortex call EDA Report

Data Quality Findings (Notebook 1)

1. Data loading

We have 2160 usable EEG trials after matching signal files with their labels. 240 extra label rows were removed to make the numbers match, and this was already checked as safe.

2. Class balance

The dataset is almost perfectly balanced: 1082 LEFT trials and 1078 RIGHT trials (about 50/50). No issue here, nothing to fix.

3. Trial length

Every single trial has exactly 2500 rows. All 2160 files are the same length. No inconsistency at all.

4. Sampling rate problem (still unresolved)

The Time column inside each file does not tick at a steady rate. The gaps between readings jump around a lot instead of being even, which means we can't safely calculate the real sampling rate from this column. We are currently using a placeholder guess (250 Hz) just to keep testing code, but this must be confirmed with the instructor before we finalize any filtering settings.

5. Missing values

Checked all 2160 trials for missing data (NaNs). Found zero. The dataset is fully complete, nothing missing anywhere.

6. Noisy/duplicate channel issue (C3 vs C4)

When we compared C3 and C4 signals directly, about 46% of trials showed the two channels behaving almost identically, which is unusual since these two channels are supposed to differ (that difference is literally what tells LEFT from RIGHT). We tested if this was a real hardware problem or just raw unfiltered noise, and it looks like it's mostly caused by low-frequency drift dominating the raw signal — something a proper filter later should mostly fix. Not confirmed as a hardware fault. The high-correlation group also splits evenly between LEFT and RIGHT (495 vs 492), so it's not biased toward one class.

7. Flat / dead channels

Checked if any channel was completely flat (broken electrode). Found zero flat channels across the whole dataset. This is good news.

8. Extreme variability (artifact check #1)

Checked each channel in each trial for unusually wild swings compared to its normal range. Found 398 flagged instances, affecting 187 unique trials (8.7% of the dataset). CZ was the most problematic channel, FZ second, then C3 and C4 were the calmest.

9. Sudden jump check (artifact check #2)

Checked each trial for one big sudden spike anywhere in the recording (like an electrode losing contact for a moment). Found 341 flagged instances. Same pattern: CZ worst, FZ second, C3 and C4 calmer.

10. High-confidence bad trials

57 trials were flagged by BOTH artifact checks above — meaning two separate, independent tests agree something is wrong with these specific recordings. This 57-trial group is our strongest, most trustworthy list of "likely bad" trials. It also splits evenly across LEFT (29) and RIGHT (28), so it's not secretly hurting one class more than the other.

Overall picture: the dataset is clean in the basics (balanced, complete, consistent length, no flat channels), but has two real open issues — the unreliable sampling rate that needs instructor confirmation, and a small, well-identified group of noisy trials (highest confidence: 57 trials) that should likely be excluded or closely reviewed once we start real preprocessing next week.

Three decisions with Outliers :

Option 1 — Play it safe. Use the $3.0\times$ cutoff you already tested. It only removes ~9% of your data (leaves ~1972 trials), so you're not losing much.

Option 2 — Skip the outlier rule entirely. Instead, just throw out the 57 trials that both methods flagged as bad. That's only 2.6% of your data—a much smaller, cleaner cut.

Option 3 — Clean up the data first. Run a band-pass filter to remove low-frequency drift, then redo the rejection curve. It might give you a clear "elbow" to justify your cutoff, instead of guessing.

Probably we will go with option two

PSD / ERD Findings (Notebook 2)

1. Rejection rule — quantifying thresholds

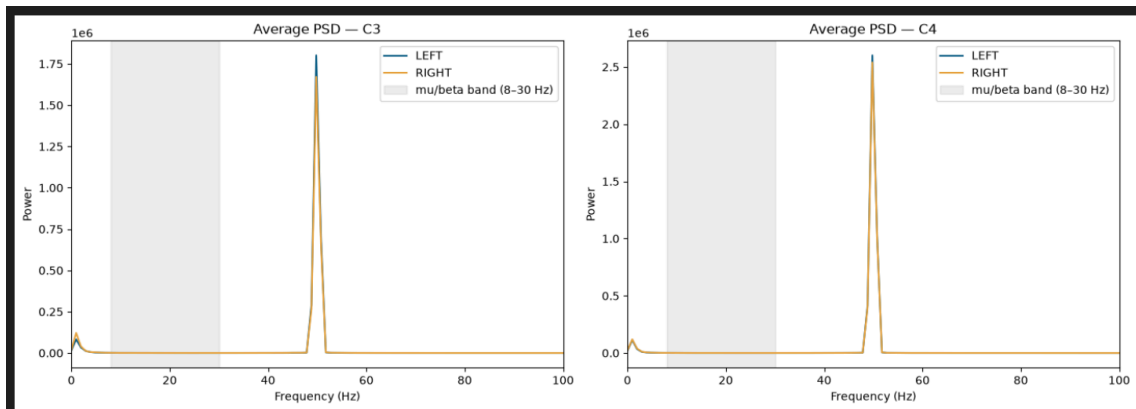
Tested IQR strictness from $1.5\times$ to $4.0\times$. Rejection rate moves smoothly from 10.9% down to 7.8% as the threshold loosens, with no clear "elbow" — meaning there's no single obvious cutoff point where bad and good trials naturally separate.

2. Final rejection list

This notebook recomputed the cross-validated rejection list (trials flagged by both peak-to-peak AND max-jump) and got **165 trials (7.6%)** — This rejected list is saved in a csv file and it's going to be cleaned soon

3. Raw PSD, LEFT vs RIGHT, C3/C4

Confirmed what we suspected: raw signal is dominated by low-frequency drift (0-3 Hz), completely masking the 8-30 Hz mu/beta band. LEFT and RIGHT look identical in raw form — not because there's no real difference, but because the drift hides it.



4. Filtered PSD, LEFT vs RIGHT, C3/C4 — real separation found

After applying an 8-30 Hz band-pass filter, a clear, visible LEFT vs RIGHT separation appears on both C3 and C4. RIGHT consistently sits above LEFT across most of the mu/beta range, especially around the 10-12 Hz mu peak.

